

Narrated PowerPoint Presentations

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Neuroendocrine Hormones of the Hypothalamus

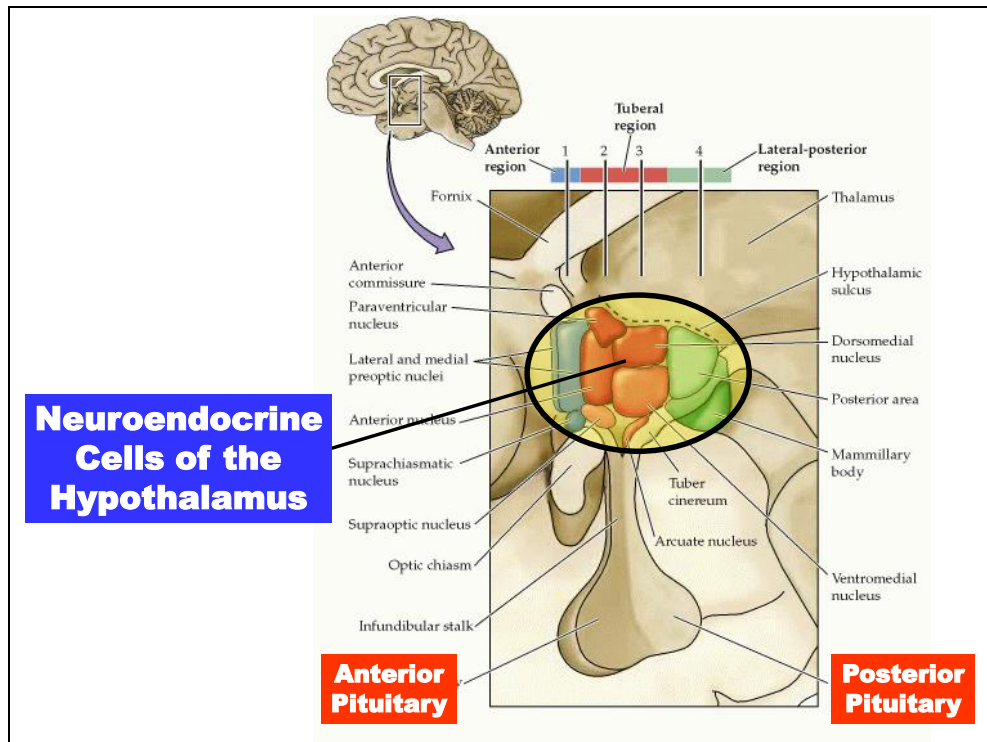
Discussion of specific hormones begins here.

Hypothalamic Hormone Properties

- **Classification:** All Peptides (except **Dopamine**)
- **Biosynthesis:** Hypothalamic Peptidergic Neurons
- **Storage:** **Vesicles** (Neuron Terminals)
- **Secretion:** **Exocytosis: Rapid:** Portal or Systemic Circulation
- **Transport:** **Dissolved in Plasma** (pg-ng/ml)
- **Metabolism:** **Proteolysis:** Short Half-Lives (mins.)
- **Mechanism:** **G-Protein Coupled:** **cAMP** or **DAG-IP3**
- **Actions:** **Rapid** (secs.-mins.): Ant. Pituitary or Peripheral Tissue
- **Regulation:** **Negative Feedback** (Hormone/Metabolite/Nerve)
- **Disorders:** **Hypo- or Hyper-Secretion** (Many & Diverse)

All hypothalamic hormones possess basic properties typical of the **Protein/Peptide (or Catecholamine)** structural class of hormones as described in greater detail in the Overview section.

Highlights are summarized above as review.

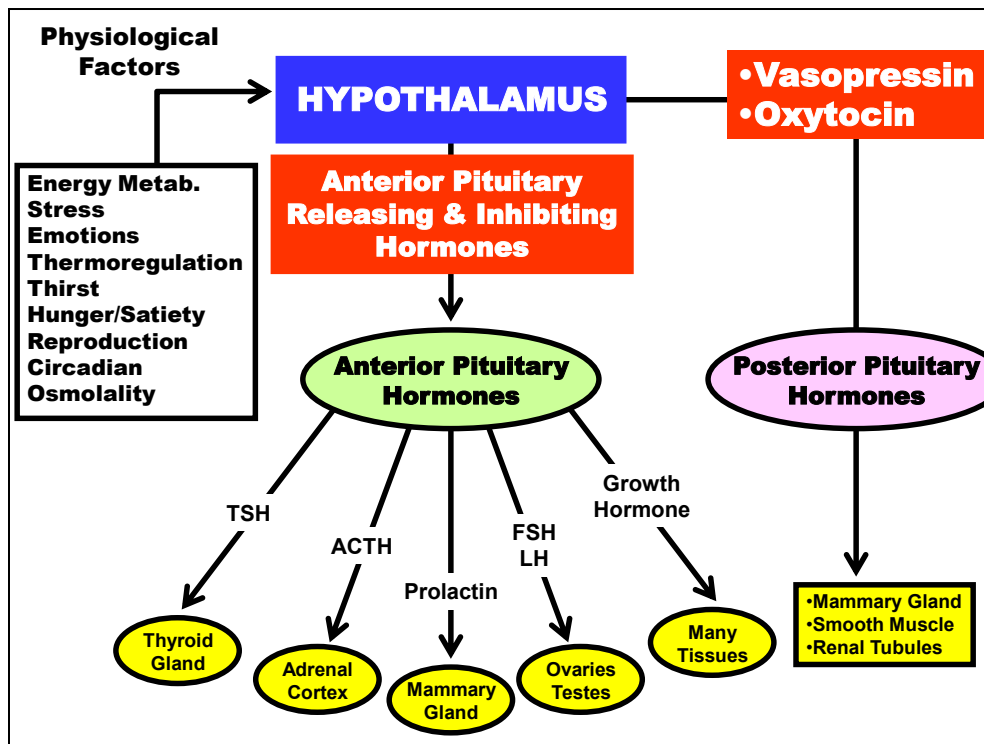


The Hypothalamus is located near the base of the brain (*top left*) and exists in close anatomical association with the Anterior & Posterior Pituitary Glands.

The hypothalamus is composed of neural tissue. Some neurons are known to secrete endocrine hormones and are termed **Neuroendocrine Cells**.

As outlined in the next figure, the endocrine functions of the hypothalamus and pituitary gland, which also functions as an endocrine gland are also closely integrated.

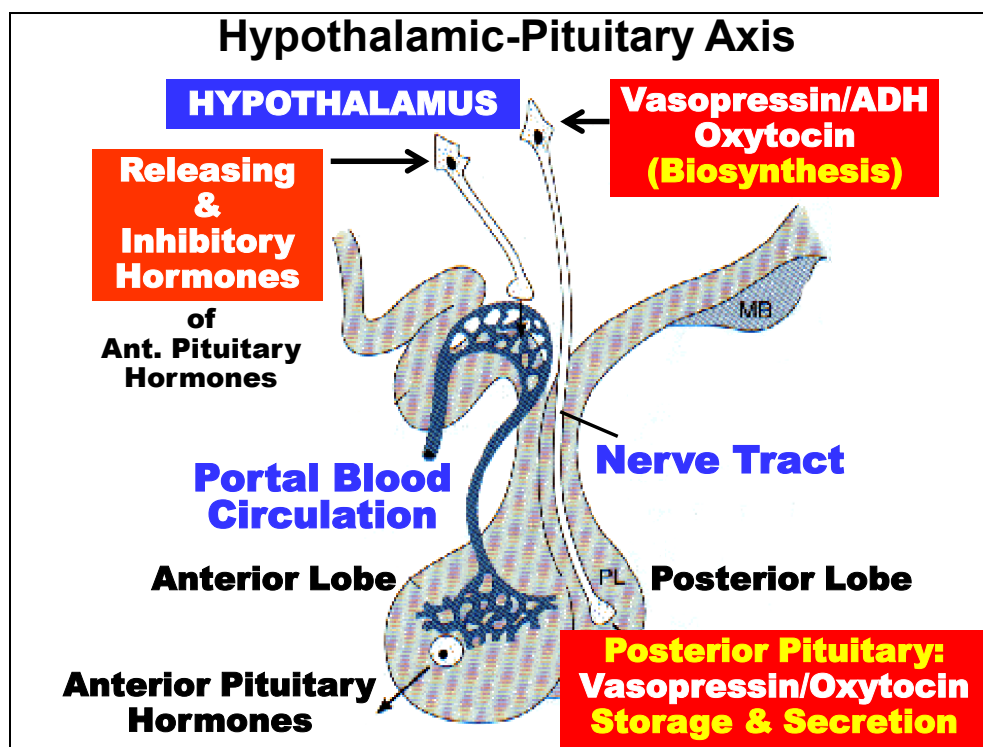
Together, they form what is known as the **Hypothalamic-Pituitary Endocrine Axis**



Certain neurons of the **Hypothalamus** respond to a variety of Physiological Factors (top left) by secreting Hormones that are divided into two groups:

1. Hypothalamic Releasing & Inhibiting Hormones functions to regulate the secretion of specific hormones produced by the Anterior Pituitary Gland. Anterior Pituitary Hormones function to regulate secretion from other Endocrine Glands (e.g. Thyroid, Adrenal, Ovaries, Testes) or the functions of other tissues involved in maintaining Homeostasis of parameters related to the signaling hypothalamic Physiological Factors.

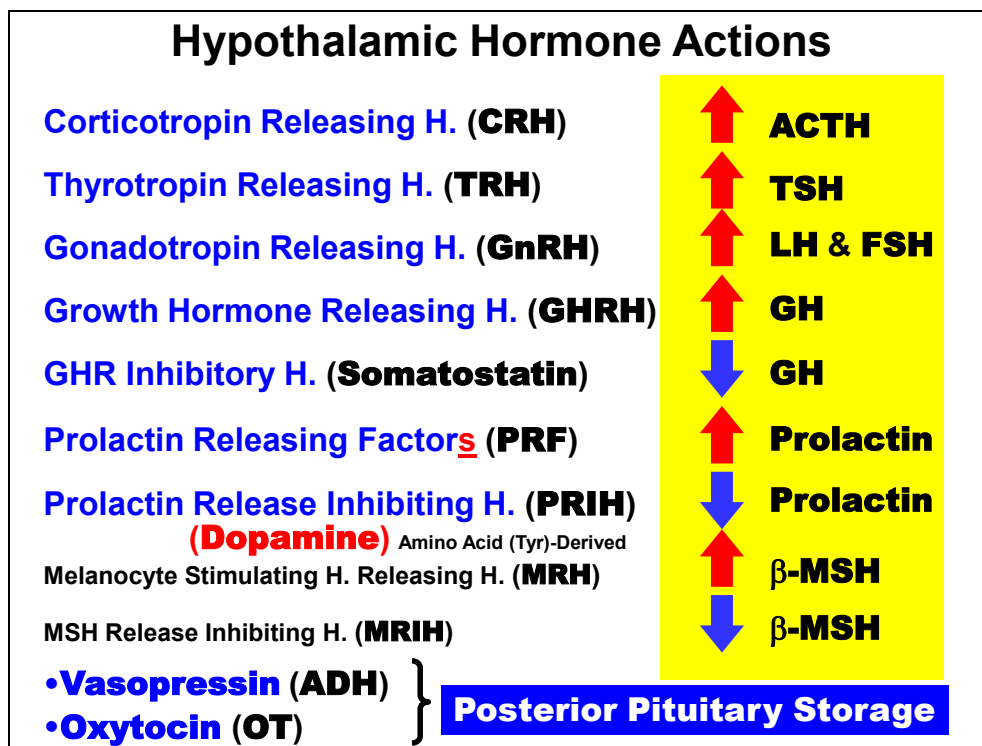
2. Two other Hormones (Vasopressin & Oxytocin) are **Biosynthesized** by **Hypothalamic Neuroendocrine Cells**, but **Stored** in the **Posterior Pituitary Gland** prior to secretion into the systemic circulation.



An important neuroendocrine relationship exists anatomically between the **Hypothalamus** and **Pituitary Glands**. Note that neurons of the hypothalamus (*top center*) communicate via Distinct Routes to the **Posterior Lobe** (*right*) and **Anterior Lobe** (*left*) of the Pituitary Gland.

Neuroendocrine cells of the **Hypothalamus** that biosynthesize & secrete **Anterior Pituitary Releasing & Inhibiting Hormones** affect cells of the Anterior Lobe via neuronal axons terminating near a hypothalamic-hypophyseal **Portal Blood Circulation** between the hypothalamus and anterior lobe. Secretion of hypothalamic hormones into this local circulation allows the channeling of these hormones directly to the anterior lobe at high relative concentrations.

Other hormones biosynthesized by the hypothalamus (**Vasopressin/ADH (Antidiuretic Hormone) & Oxytocin**), are transported down long axons that extend along a hypothalamic-hypophyseal **Nerve Tract** that terminate within the **Posterior Lobe**. Hormones are stored within nerve terminals until secreted (**Posterior Pituitary Hormone Storage**). Thus, a direct nervous connection exists between the hypothalamus & posterior lobe.



Hypothalamic Releasing & Inhibiting Hormones (*top nine*) and their specific action on anterior pituitary hormone secretion (*right box*) are listed above. Melanocyte Stimulating Hormone Releasing (**MRH**) and Inhibitory (**MRIH**) Hormones will not be discussed in detail. However, know that these two hormones are synthesized and secreted from the hypothalamus.

Note that some hormones stimulate ↑ (e.g. **PRF**) whereas others inhibit ↓ (e.g. **PRIH**) the same anterior pituitary hormone (e.g. **Prolactin**).

Physiological shifts in the balance between these antagonistic controls accounts, in part, for the pulsatile secretion of most hormones. Also note that inhibitory counterparts do NOT exist for others (e.g. **CRH**, **TRH**). This is because the peripheral hormones that they ultimately control (cortisol, thyroid hormones) act as inhibitors of hypothalamic &/or pituitary trophic hormones via specific feedback mechanisms described earlier.

Hypothalamic neurons are also responsible for the biosynthesis of **Vasopressin (ADH)** and **Oxytocin** (*bottom*), which are termed **Posterior Pituitary Hormones** because of their storage and secretion from the gland. Functions will be discussed later in the section.

Narrated PowerPoint Presentations

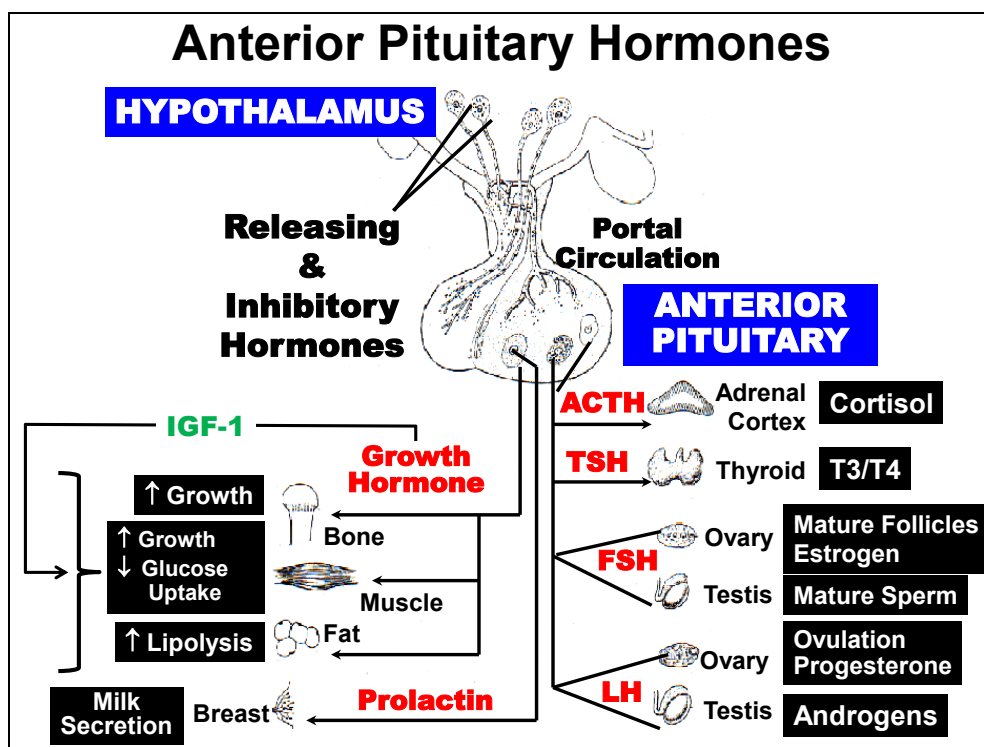
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Anterior Pituitary Hormone Properties

- **Classification:** All **Peptides/Proteins**
- **Biosynthesis:** **Corticotrophs, Thyrotrophs, Lactotrophs, etc.**
- **Storage:** **Vesicles**
- **Secretion:** **Exocytosis: Rapid:** **Systemic Circulation**
- **Transport:** **Dissolved in Plasma** (pg-ng/ml)
- **Metabolism:** **Proteolysis:** **Short Half-Lives** (mins.)
- **Mechanism:** **G-Protein/Tyrosine Kinase**
- **Actions:** **Rapid** (secs.-mins.): **Peripheral Glands/Tissues**
- **Regulation:** **Negative Feedback** (Hormone/Metabolite/Nerve)
- **Disorders:** **Hypo- or Hyper-Secretion** (Many & Diverse)

All anterior pituitary hormones possess basic properties typical of the **Protein/Peptide** structural class of hormones as described in greater detail in the Overview section.

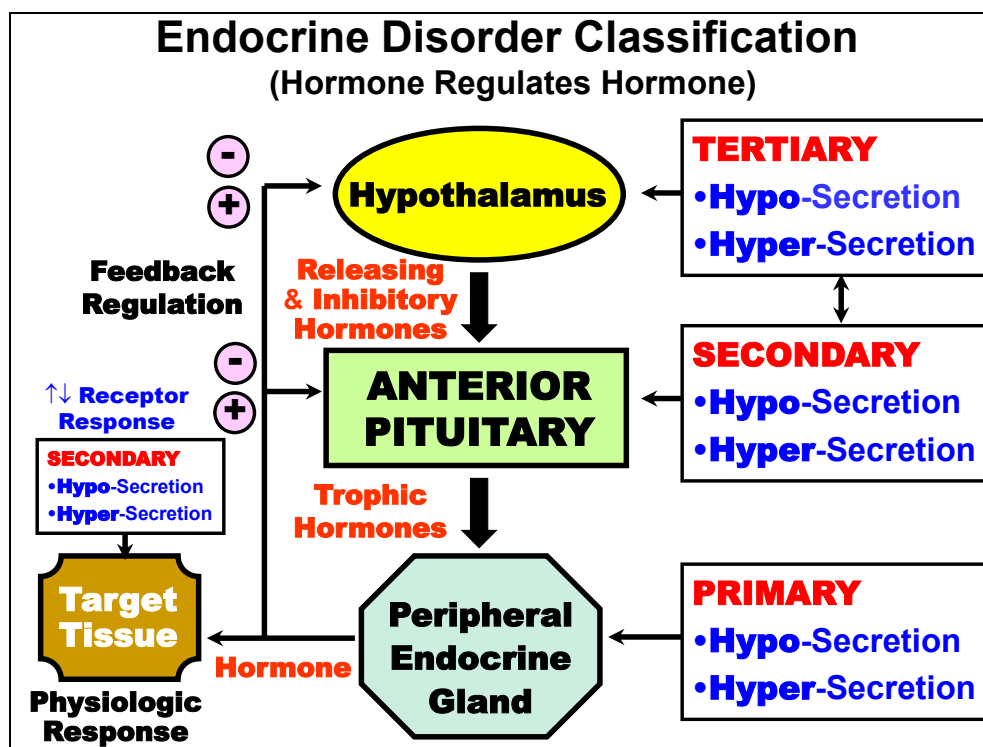
Highlights are summarized above as review.



Anterior Pituitary hormone production and secretion is directly regulated by the actions of **Releasing & Inhibitory Hormones** of the **Hypothalamus**.

The **Anterior Pituitary Gland** produces and secretes six major hormones (**TSH, ACTH, GH, PRL, LH & FSH**). Secreted hormones are summarized above along with their primary **Target Tissue(s)** and **Major Actions**.

Note the diversity of **Target Tissues** regulated by the pituitary hormones.

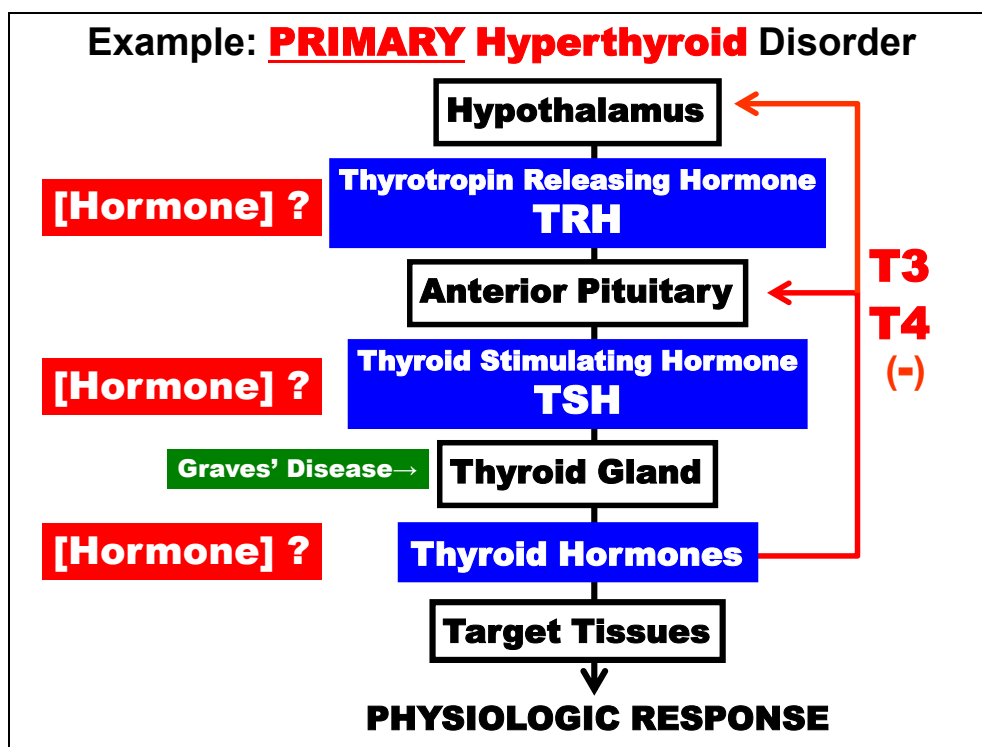


Summarized above is the basic Hypothalamus-Anterior Pituitary hormonal axis that controls Peripheral Endocrine Gland hormone secretions which, in turn, regulate Target Tissue functions (i.e. Physiologic Response). Secretion disorders of hormones controlled by this type of feedback regulatory loop are conventionally classified according to the origin or **Site of Impairment** along the axis.

Abnormalities directly affecting the **Peripheral Endocrine Organ** (e.g. thyroid or adrenal gland) are termed **PRIMARY Hyper-Secretion** or **Hypo-Secretion** disorders. e.g. Graves' Disease (a form of hyperthyroidism) is caused by immunological hyperactivity of the *Thyroid Gland* itself and is therefore termed (Primary) 1° Hyperthyroidism.

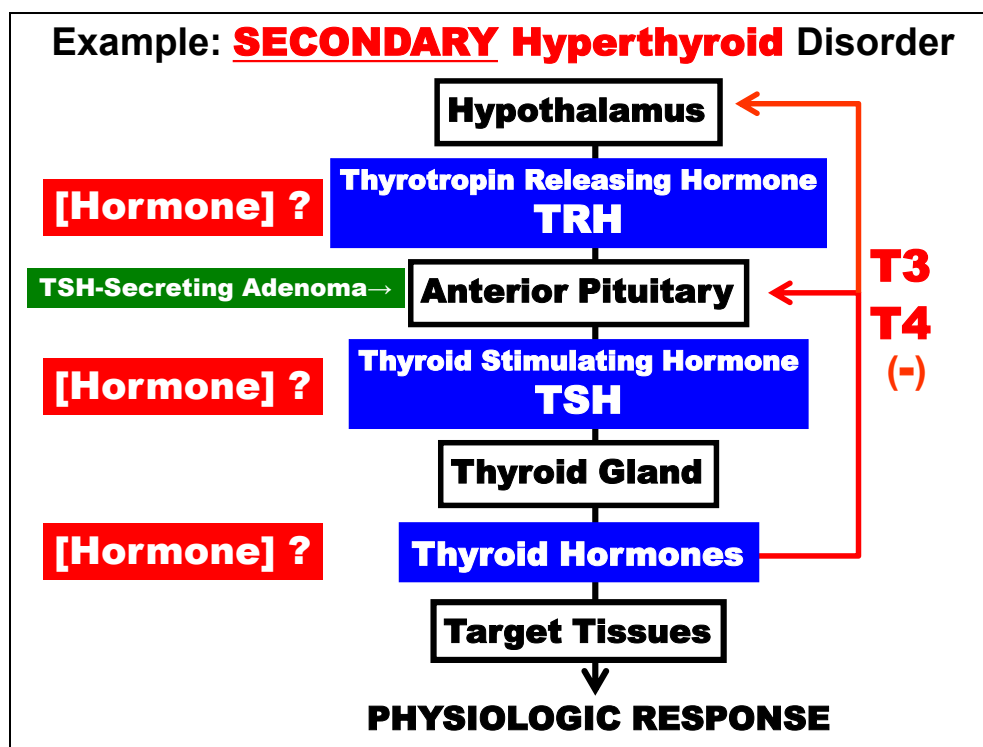
Conversely, abnormalities directly affecting the **Anterior Pituitary Gland** are termed **SECONDARY Hyper-Secretion** or **Hypo-Secretion** disorders. e.g. Hypo-Secretion of Thyroid Hormone CAUSED by destruction of TSH secreting pituitary thyrotrophs is classified as 2° Hypothyroidism. Note that conditions impairing (↑↓) **Target Tissue Response** (bottom left) to hormone action are also conventionally classified as **SECONDARY Hyper-Secretion** or **Hypo-Secretion** disorders occurring "secondary" to target tissue responsiveness, which is typically related to a **Hormone Receptor** disorders.

Finally, abnormalities of the **Hypothalamus** are formally termed **TERTIARY Hyper-Secretion** or **Hypo-Function** disorders.



What are the expected relative levels (i.e. increased or decreased) of the hypothalamic, anterior pituitary and peripheral hormones resulting from a **PRIMARY** Hypothalamic-Anterior Pituitary Axis Disorder?

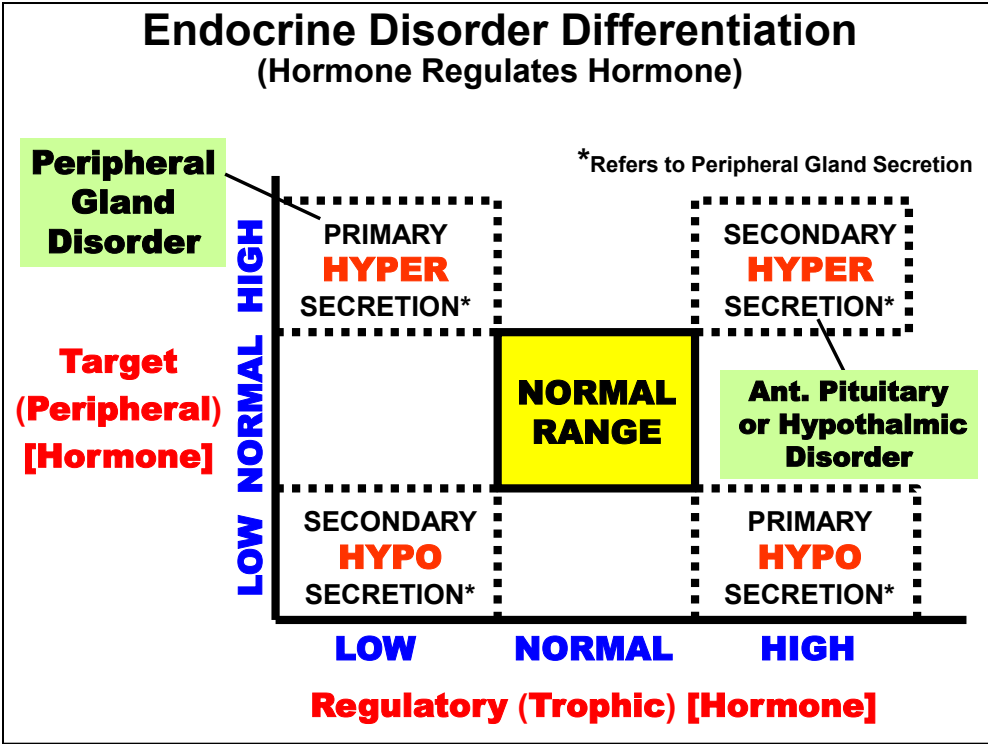
In the case of a Primary Hyperthyroid condition, predict how the plasma hormone levels ([Hormone]) of TRH, TSH, and the Thyroid Hormones would be affected relative to normal levels.



What are the expected relative levels (i.e. increased or decreased) of the hypothalamic, anterior pituitary and peripheral hormones resulting from a **SECONDARY** Hypothalamic-Anterior Pituitary Axis Disorder?

In the case of a Secondary Hyperthyroid condition, predict how the plasma hormone levels ([Hormone]) of TRH, TSH, and the Thyroid Hormones would be affected relative to normal levels and compared to the Primary Hyperthyroid condition.

Finally, rationalize how the above could be distinguished from the outcome of a **Tertiary** hyperthyroid disorder.



In instances where hormones are regulated by other hormones, **Primary vs. Secondary Disorders** can be differentiated based on laboratory measurements of the **Regulatory (Trophic) [Hormone]** (*x-axis*) and its regulated **Target (Peripheral) [Hormone]** (*y-axis*).

Hypothalamic hormone levels (**Tertiary**) may also be considered in the differential diagnosis but is not illustrated here.

Predictions are based on **Negative Feedback Inhibition** pathways that regulate **Regulatory (Trophic) [Hormone]** relative to **Target (Peripheral) [Hormone]** as shown above.

For example, Graves' Disease patients typically possess greatly increased plasma **T3 & T4** levels (**HIGH Target [Hormone]**). However, **TSH** is abnormally decreased (**LOW Regulatory [Hormone]**) because the elevated T3 & T4 levels suppress thyrotroph secretion through Negative Feedback Inhibition.

As indicated by the graph above, this is diagnostic of a (*top left*) **Primary HYPER-Secretion** (i.e. **1° Hyperthyroidism**), which was one of the examples presented earlier.

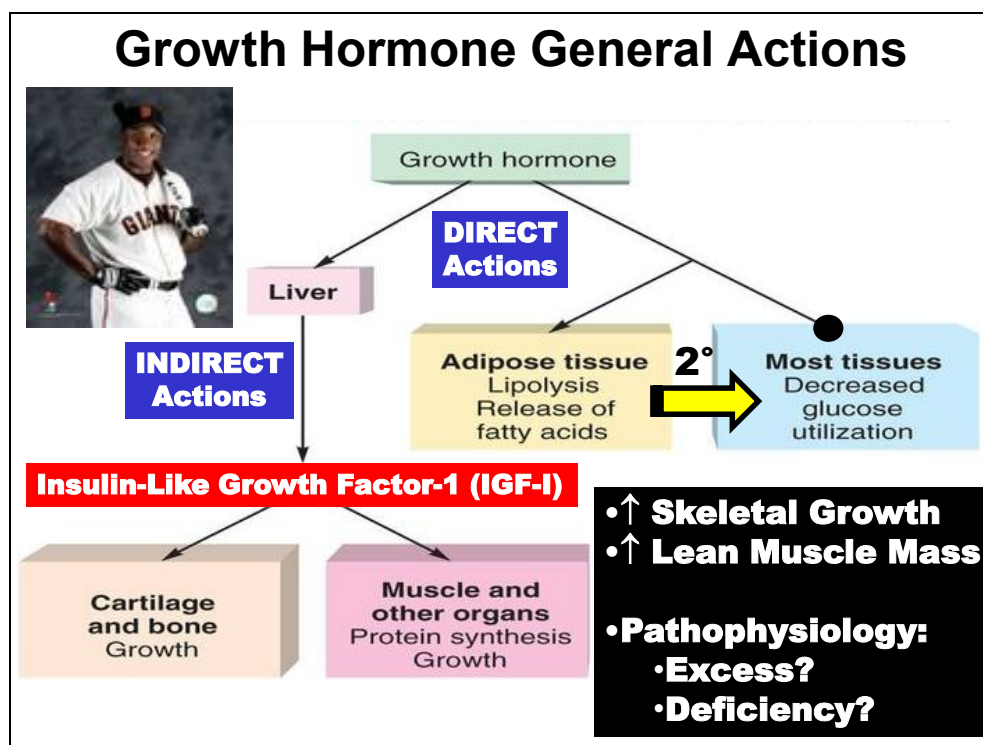
GROWTH HORMONE

Growth Hormone Properties

- **Classification:** **Protein**
- **Biosynthesis:** **Anterior Pituitary Somatotrophs**
- **Storage:** **Yes (Secretory Vesicles)**
- **Secretion:** **Exocytosis: Rapid**
- **Transport:** **50% Bound: GHBP**
- **Metabolism:** **Proteolysis: “Long” Half-Life (6-20 min)**
- **Mechanism:** **Membrane Receptor: Tyrosine Kinase**
- **Actions:** **Muscle/Bone/Fat/Liver (Direct/Indirect-IGF-1)**
- **Regulation:** **Negative Feedback: GHRH/Somatostatin**
- **Disorders:** **Hypo- or Hyper-Secretion (Dwarfism/Acromegaly/Gigantism)**

Growth Hormone endocrine properties

Highlights are summarized above.



Skeletal Growth. The principle action of **Growth Hormone** is to stimulate Skeletal Growth. GH exerts its actions on many processes related to **Bone** (Osteogenesis) & **Cartilage** (Chondrogenesis) **Growth**. Thus, GH plays an important role in adolescent growth.

Metabolic Actions. GH stimulates (↑) Protein Synthesis, which promotes ↑ **Muscle Mass** & other **Soft Tissue (Organ) Growth**. Specific GH actions include enhancing amino acid transport into cells, stimulating protein gene transcription & translation and suppressing protein catabolism & amino acid metabolism. GH also stimulates **Adipose Tissue** (↑) Lipolysis, which ↑ **Fat Utilization** by tissues and ↑ **Blood FFA** levels. Thus, GH has the overall effect of Increasing Lean Muscle Mass.

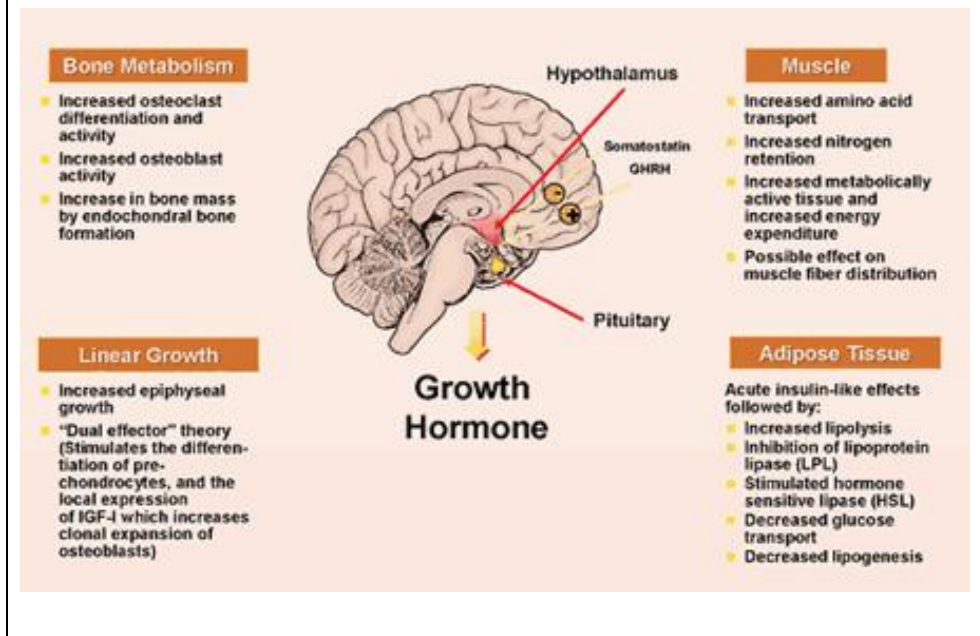
GH also decreases (↓) Glucose Utilization by most tissues. Effect is secondary (2°→) through its stimulation of lipolysis (glucose sparing effect), which decreases glucose uptake and utilization of most tissues, which promotes ↑ **Blood Glucose**. Thus, GH indirectly ↑ **Insulin Secretion**. As discussed later, chronic excess GH levels may increase the risk of developing hyperglycemia leading to hyperinsulinism and a form of “**Pituitary Diabetes Mellitus**”.

Mechanisms of Actions (diagram).

(left) Many GH actions are mediated Indirectly by a group of proteins called **Somatomedins** (**INDIRECT**). The principle somatomedin related to growth hormone actions is **Insulin-Like Growth Factor I (IGF-1)** or **Somatomedin C**. Circulating **IGF-1** is mainly secreted and stimulated by GH from the **Liver**. It is also secreted in Cartilage, Bone & Muscle in response to GH stimulation, acting as a paracrine agent.

(right) Growth Hormone can also act Directly by binding to its own (tyrosine kinase) receptor on target tissues. Stimulation of hepatic IGF-1 and metabolic actions of GH such as adipose tissue lipolysis are believed to be regulated directly by GH.

Growth Hormone Specific Actions

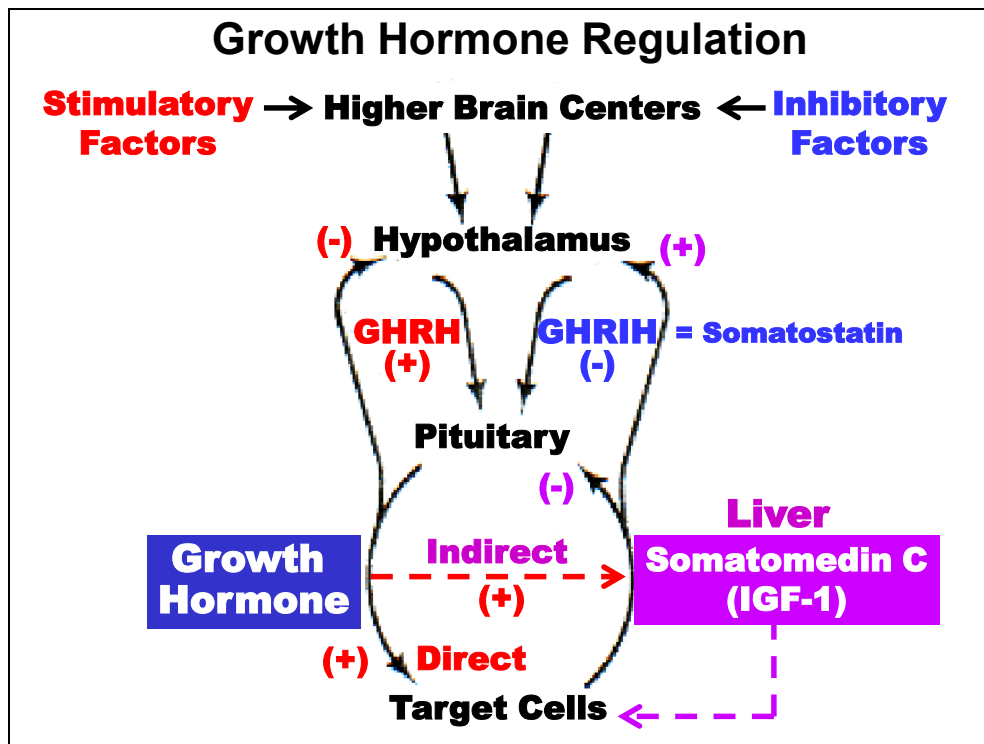


Specific cellular actions regulated by **Growth Hormone**.

No need to know details above. Only target tissues and general actions as outlined in previous figure.

Table 1 Hypothalamic & Pituitary Hormones	
Hormone	
Corticotropin Releasing H. (CRH)	Hypothalamus
Thyrotropin Releasing H. (TRH)	
Gonadotropin Releasing H. (GnRH)	
Growth Hormone Releasing H. (GHRH)	
Somatostatin (GHRIH)	
Prolactin Releasing Factors (PRF)	
Prolactin Release Inhibiting H. (PRIH)	GH Regulation
Adrenocorticotrophic H. (ACTH)	Anterior Pituitary
Thyroid Stimulating H. (TSH)	
Leutinizing Hormone (LH)	
Follicle Stimulating H. (FSH)	
Growth Hormone (GH)	
Prolactin (PRL)	
Vasopressin (ADH)	"Posterior Pituitary" (Hypothalamus)
Oxytocin (OT)	

A reminder that Growth Hormone secretion is regulated via a classic Hypothalamic-Anterior Pituitary axis involving the balance of stimulatory (GHRH) and inhibitory (GHRIH i.e. Somatostatin) hormones secreted from hypothalamic neuroendocrine cells.



Growth Hormone secretion is controlled by the balance between two antagonistic regulatory hormones of the **Hypothalamus** as diagrammed above.

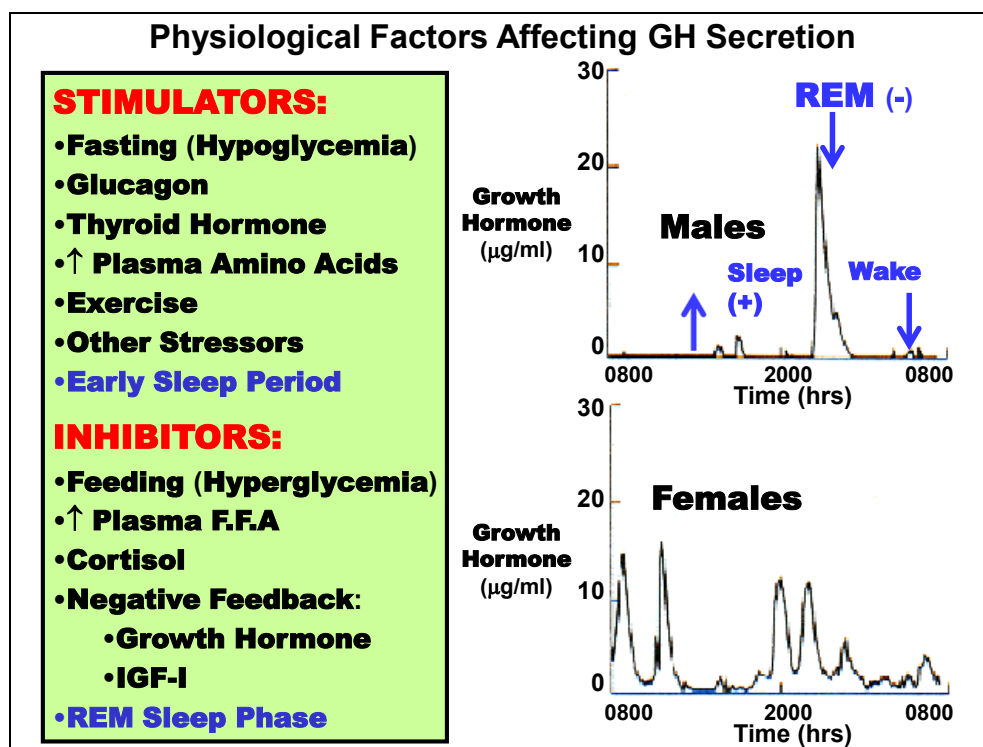
Growth Hormone Releasing Hormone (**GHRH**) stimulates (+) **Growth Hormone** secretion from the **Anterior Pituitary** (*center left*). Growth Hormone Release Inhibitory Hormone (**GHRH = Somatostatin**) inhibits (-) GH secretion from the **Anterior Pituitary** (*center right*). .

(*top left & right*) The predominant effect (GH release or inhibition) is determined by the relative influence of various physiological **Stimulatory** and/or **Inhibitory Factors**.

The *left downward arrow* indicates the **Direct** GH actions. The *dashed arrows* indicate the **Indirect** actions of GH on **Target Cells** via **Somatomedin C (IGF-I)**.

The basic **Negative Feedback** mechanism (*upward arrows*) involves **Growth Hormone** (GHRH inhibition-*left arrows*) itself and **Somatomedins-IGF-1** (GH inhibition & GHRH stimulation-*right arrows*).

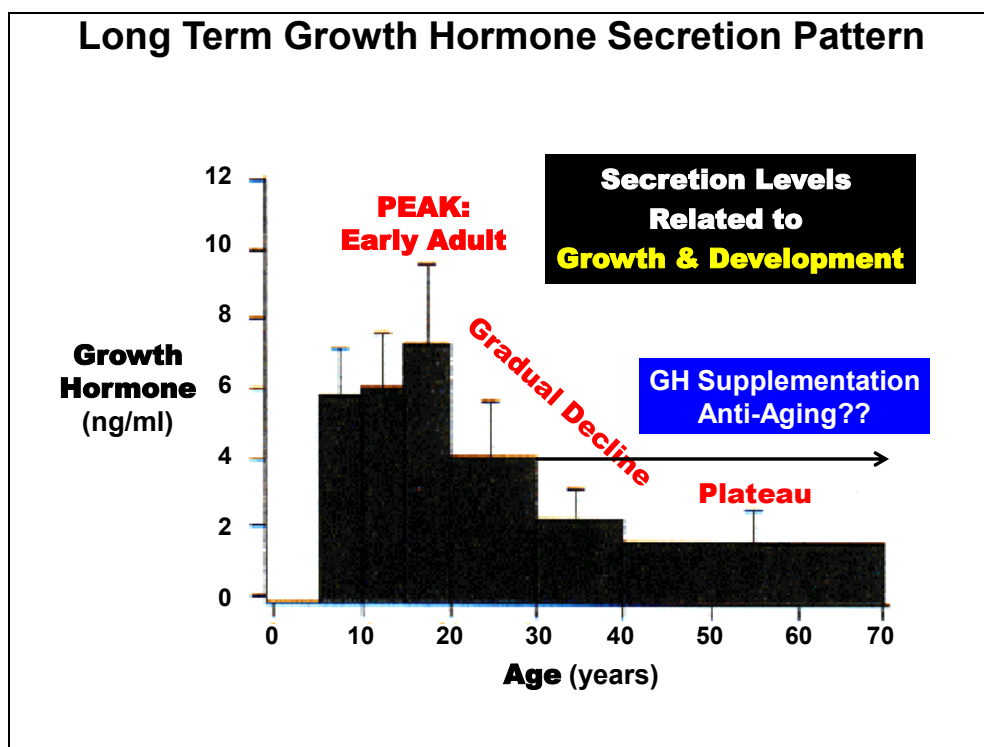
Be prepared to classify growth hormone disorders are primary, secondary, etc.



Some of the physiological factors that Stimulate or Inhibit GH secretion are listed (*left*). Note that most are consistent with GH physiological actions. For example, conditions that are related to or require increased energy substrate mobilization (**Fasting, Exercise, Glucagon**) apparently stimulate GH secretion (i.e. related to Lipolytic & Hyperglycemic actions).

The daily secretion pattern of growth hormone occurs in **Episodic Bursts**. Low level pulses occur throughout the day, with peak secretion rate usually occurring during the early **Sleep** period. This is most prominent in **Males** (*top left*) as GH secretion is typically more irregular in **Females** (*bottom left*). As indicated by the sharp decline within the sleep period, the later **REM stage inhibits (-)** GH secretion. The physiological reason for the two latter effects is unclear.

Sleep-related pulses account for most ($\approx 70\%$) of the daily GH secretion and are greater in growing infants & adolescents. Sleep-related pulses are independent of the individual's **Circadian** sleep-wake cycle (unlike cortisol). That is, GH sleep pulses apparently do NOT occur if one remains awake during their normal sleep period (e.g. midnight-8 AM), but becomes stimulated even if sleep occurs during the normal wake period (e.g., 8 AM-midnight).



GH levels are highest in growing children and teenagers. Thereafter, GH levels gradually decline to a steady-state level at around age 40.

Note peak around late teens early twenties and low plateau around fourth decade.

Because of its physiological role and metabolic effects, GH is currently touted by some as an “Anti-Aging” supplement. However, some potentially harmful side-effect may occur with GH supplementation as summarized later.

Growth Hormone (GH) Supplementation

Little is Known About the Efficacy & Safety of “Anti-Aging” Supplements

Some Hypothetical “Anti-Aging” Benefits of GH Supplementation:

SKIN - Increased skin elasticity, texture, and tightness.
ENERGY - Increased energy and emotional stability.
BONE - Improved bone strength.
SEXUAL POWER Increased sexual potency & frequency.
MUSCLE - Increased muscle strength and mass.
FAT - Decreased fat tissue.
MEMORY - Improved mental functioning and strength.
HEART - Improved cardiovascular strength & lower blood pressure.
KIDNEY - Improved kidney function.
IMMUNE SYSTEM - Improved immunity and healing.
HAIR - Improved hair texture.
CHOLESTEROL - Elevated HDL and lowered LDL.



Many products exploiting the theoretical benefits of GH supplementation are abundantly available.

Most contain only homeopathic levels of GH or are simply additives (e.g. amino acids) touted as natural GH stimulants. Some may contain small amounts of GHRH.

However, more potent formulations of GH are apparently quite easily obtained (legally or illegally) if one is willing to spend the money.

Almost all over-the-counter products are oral supplements...GH is a protein (is any actually absorbed?).

The primary clinical concern with these products is that little evidence based data is available concerning the effectiveness and safety of GH supplementation in otherwise healthy individuals. Ethical issues related to consumer financial exploitation by manufacturers is also an issue.

Growth Hormone Disorders



Acromegaly

Effects Depend on Onset

DEFICIENCY:

- **Dwarfism** (Adolescent Onset)
 - ↓ GH Secretion; ↓ IGF-1
 - ↓ GH Sensitivity
 - **Laron Dwarfism**
 - ↔ ↑ GH ; ↓ IGF-1
- **Obesity** (Adult Onset)

EXCESS:

- **Gigantism** (Adolescent Onset)
 - ↑ Bone Length
 - ↑ Bone Thickness
- **Acromegaly** (Adult Onset)
 - ↑ Bone Thickness:
 - Nose, Head, Hands & Feet
- **Organomegaly**
- **“Pituitary” Diabetes**

Growth Hormone Deficiency. Results most prominently in **Short Stature (Dwarfism)** if developed BEFORE PUBERTY due to decreased capacity to promote cartilage and long bone growth. Skeletal growth deficiencies in adults are usually minimal. GH deficiency (both adolescent & adult) is also characterized by **Obesity** (decreased lipolytic activity). There are two basic causes of GH deficiency:

Decreased Growth Hormone Secretion. Hypothalamic impairments fail to adequately stimulate GH secretion (GHRH) from the anterior pituitary. Pituitary Impairments result in direct deficiencies in GH production and/or secretion. Pituitary & hypothalamic insufficiencies are both responsive to growth hormone therapy. To differentiate between hypothalamic vs. pituitary impairments clinically, the GH secretion response to injection of exogenous GHRH is increased in hypothalamic disease but remains suppressed in pituitary disease.

Primary deficiencies in GH secretion also result in deficient Somatomedin C (**IGF-I**) secretion. Patients possess LOW GH levels and LOW IGF-1 levels.

Growth Hormone Insensitivity of Target Tissues (Laron Dwarfism). Deficient or impaired **Growth Hormone Receptors** on target cells render tissues unresponsive or “**Insensitive**” to GH stimulation. Typically, GH levels are sometimes **Normal** or even **High** despite deficient target tissue response. However, **Somatomedin (IGF-I)** secretion is usually impaired due to GH receptor insensitivity. Thus, patients are distinguished Normal-HIGH GH levels, but LOW IGF-1 levels.

Growth Hormone Excess. Most common cause is **Anterior Pituitary Adenoma**.

If long bone epiphyses have **NOT Closed** (i.e. adolescent onset), excess GH stimulates both Linear & Circumferential bone growth. If the epiphyses have already **Closed** (i.e. adult onset), excess GH stimulates mainly **Circumferential** bone growth.

Thus, GH excess Before Puberty causes a condition termed **Gigantism**, resulting mainly from Excess Linear Growth of long bones. Individuals may reach a height in excess of 8 feet tall. Growth hormone excess after puberty or in adults causes a condition termed **Acromegaly** and is characterized in adults by Increased Bone Thickness, typically those of the hands, feet and cranium along with Organomegaly. Progression of condition is shown above.

A condition termed “**Pituitary Diabetes**” is a common manifestation of excess GH (pre- & post- puberty). When GH becomes chronically elevated, the metabolic effects of GH on lipolysis and glucose uptake result in an abnormally high blood glucose level. The chronic and excessive stimulation of insulin results in “**insulin insensitive**” tissues, which further impairs plasma glucose uptake ability despite high insulin levels (**Type II diabetes**). If untreated, this condition eventually leads to damage or “**burn-out**” (fatigue) of pancreatic β -islet cells and insulin insufficiency (analogous to **Type I diabetes**).

These patients typically possess **HIGH** GH levels and **HIGH** IGF-1 levels.

Laron Dwarfism Relationship to Disease Protection/Longevity?



<https://www.discovermagazine.com/health/rare-form-of-dwarfism-protects-against-cancer>

FYI-only



Images of Pre-Pubertal Growth Hormone Deficiency (*left*) and Excess (*right*).

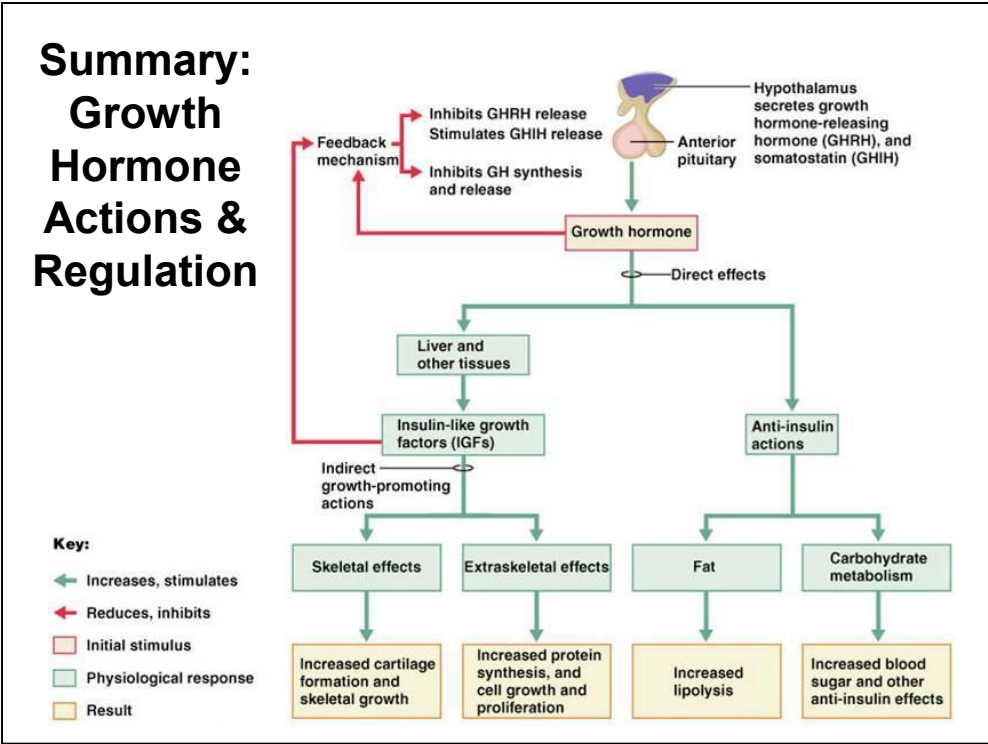


Andre the Giant

7' 4"
500 lbs

At 7'4" and 500 pounds, Andre the Giant was born Andre Rene Roussimoff in Grenoble, France on May 19, 1946. His parents and four siblings were of average size. Andre, however, suffered from Gigantism, a disease that results in an over abundance of growth hormones. This disease caused Andre's body to continue growing his whole life; by the time he was 17 he stood 6'7".

Sadly, over the years the effects of this disorder wore down his body. Eventually his immense size was just too much for his heart, and Andre the Giant died in Paris, France in his hotel room on January 27, 1993.



Summary of Actions and Regulation of Growth Hormone

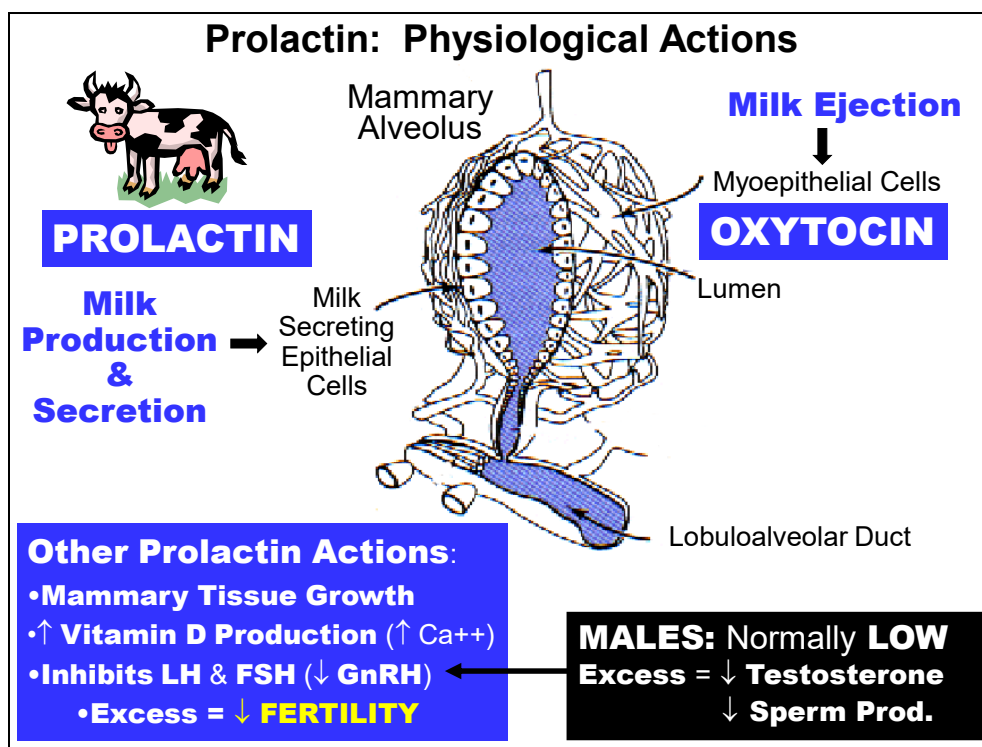
PROLACTIN

Prolactin Properties

- **Classification: Protein**
- **Biosynthesis: Anterior Pituitary Lactotrophs**
- **Storage: Yes (Secretory Vesicles)**
- **Secretion: Exocytosis: Rapid**
- **Transport: Dissolved**
- **Metabolism: Proteolysis (Half-Life: Minutes)**
- **Mechanism: Membrane Receptor: Tyrosine Kinase**
- **Actions: Mammary Glands**
- **Regulation: PRFs/PRIH (Dominant) Breast Suckling**
- **Disorders: Hypo-/Hyper-Secretion (Prolactinoma)**

Prolactin endocrine properties

Highlights are summarized above.



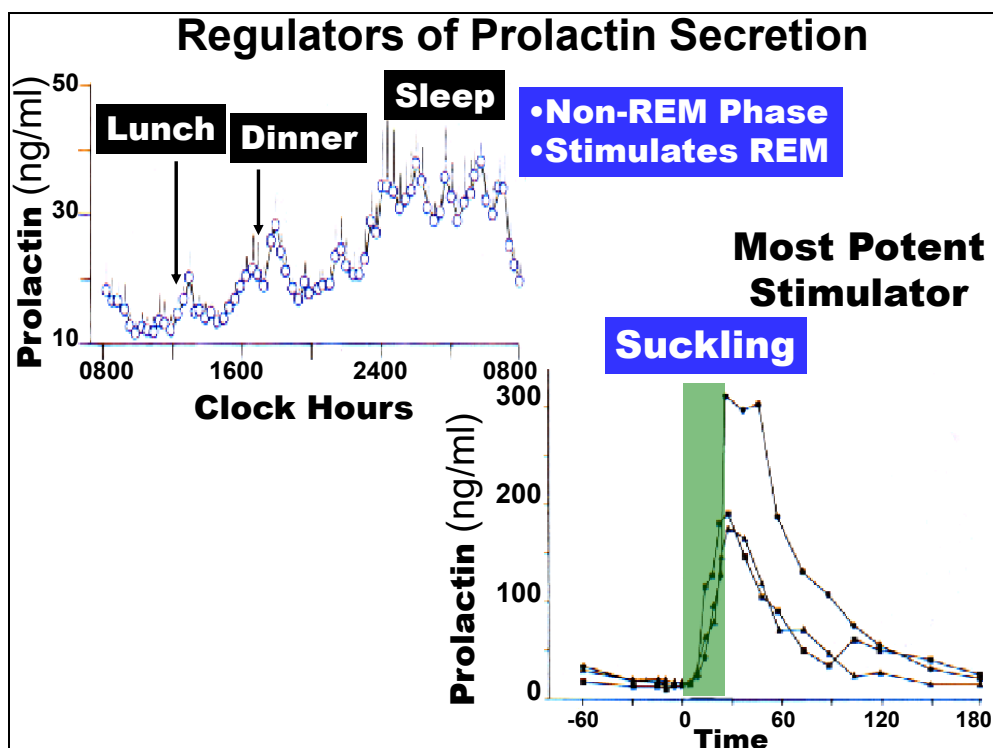
Lactation. The major physiological functions of **Prolactin (PRL)** are related to **Milk Production & Secretion** in post-partum (lactating) females. PRL stimulates the production of Milk Proteins such as casein & lactalbumin by acting on Milk Secreting Epithelial Cells lining the inner wall of Mammary Alveoli. PRL actions occur in concert with the posterior pituitary hormone **Oxytocin**, which causes **Milk Ejection** (Let-Down) by stimulating contraction of Myoepithelial Cells surrounding Mammary Alveoli.

Other Actions (*bottom left text box*) PRL also stimulates **Mammary Tissue Growth** in anticipation of lactation. Complementary to its role in milk production, PRL promotes renal (↑) **Vitamin D** production, which increases Ca⁺⁺ absorption from the GI tract.

Finally, PRL has an important inhibitory effect on the anterior pituitary gonadotrophs **LH & FSH** probably via hypothalamic **GnRH Suppression** (↓). These hormones play central roles in the ovulatory cycle and male reproductive functions. Excess PRL commonly delays resumption of ovulation in lactating females, acting as a natural, but unreliable birth control method. However, hypersecretion in non-lactating females may decrease fertility. The most common example of prolactin hypersecretion is the development of a hypersecreting anterior pituitary **Prolactinoma** (adenoma).

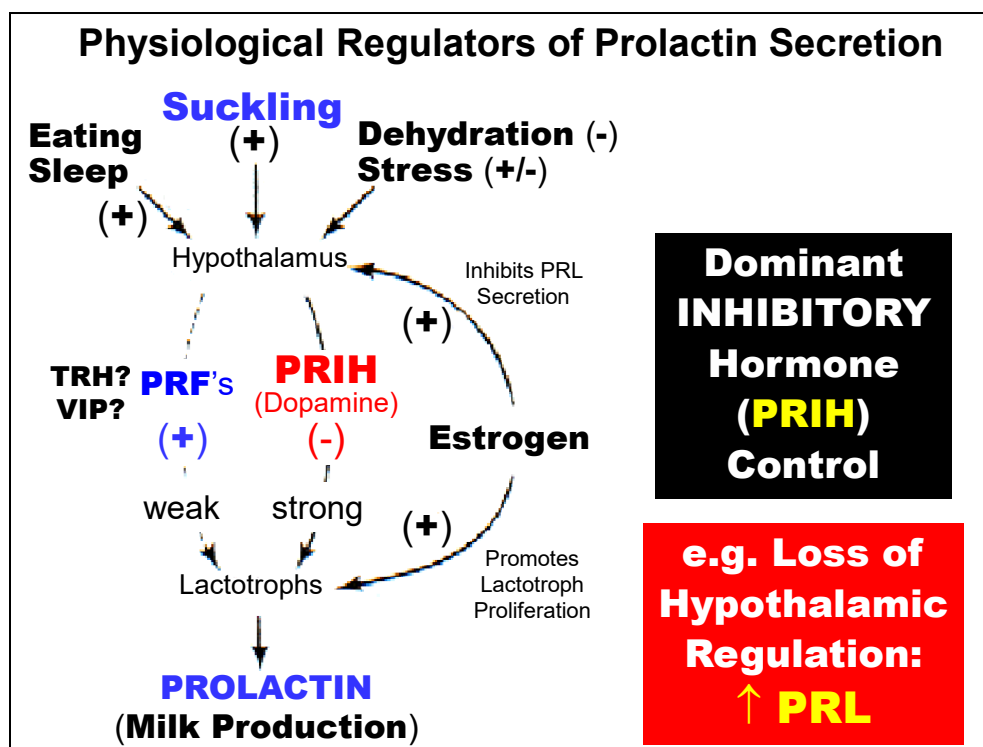
In males (*bottom right*), PRL secretion is normally low. Excess PRL suppresses

Testosterone & Sperm Production through inhibitory effects on GnRH, also acting to decrease male fertility.



(top left) Increased basal PRL secretion occurs in a daily Episodic Pattern in response to **Eating** (Lunch & Dinner) and during **Sleep**.

(top right) **Suckling** of the breast nipple during lactation is the most potent PRL stimulatory factor. Compare the 5-15x > plasma level of PRL (y-axis) induced after Suckling (*shaded area*) relative to levels stimulated during Sleep or Eating (*top left graph*). Suckling stimulated pulses are superimposed on other physiologically stimulated pulses.



In contrast to other anterior pituitary hormones, **PRL** is under **Dominant Inhibitory Hormone Control** by **PRIH (Dopamine)**. Thus, **disorders that disrupt the action of PRIH results in HYPERSECRETION of PRL.**

For example, obstruction of the hypothalamic-anterior pituitary portal circulation result sin decreased secretion of all anterior pituitary hormones, except PRL, which increases.

The **most important physiological stimulatory mechanism for PRL secretion is mechanical Suckling by infant during breast feeding.**

When Suckling (*top*) is initiated, an afferent nervous signal is transmitted from the nipple to neurons in the Hypothalamus, which stimulates (+) Prolactin by inhibiting (-) **PRIH-Dopamine** but also by stimulating PRF's although a weaker mechanism. Proposed candidates for PRFs include Thyrotropin Releasing Hormone (TRH) and Vasoactive Inhibitory Hormone VIP). Neither PRL nor Milk Production act as negative feedback products. PRL secretion ceases only when Suckling Stops.

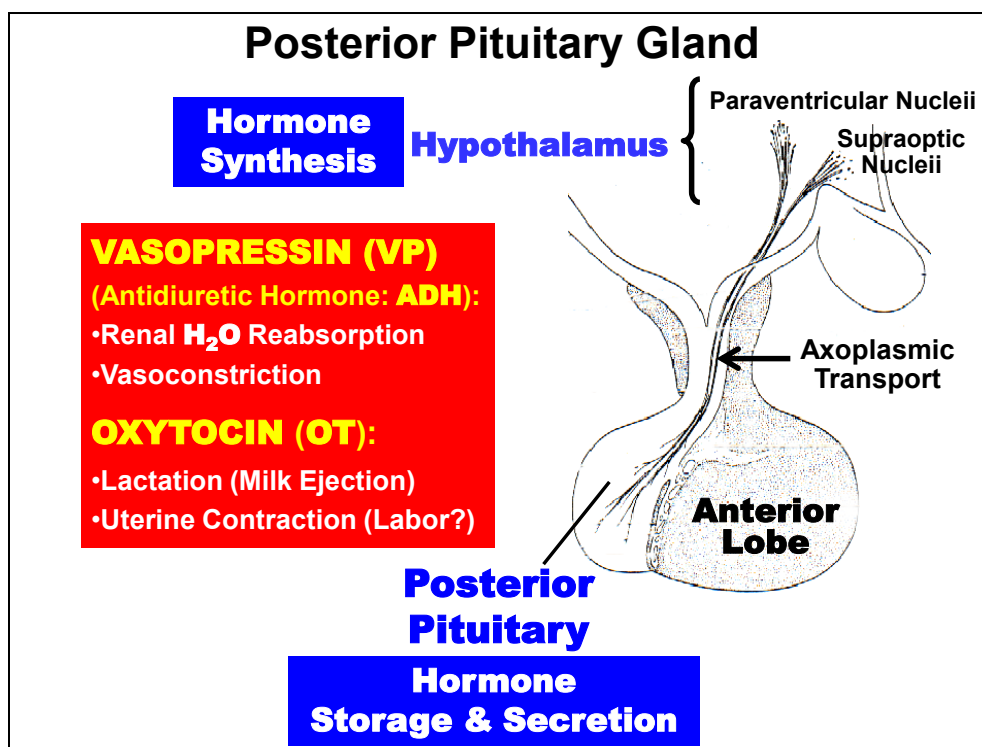
Other regulators include Estrogen (*right*), which indirectly promotes (+) milk secretion response to PRL through stimulation of Lactotrophs proliferation (*downward arrow*). However, estrogen also inhibits PRL and milk secretion thorough stimulation (+) of PRIH (*upward arrow*). This mechanism apparently suppresses PRL release in non-pregnant women and during gestation, but allows increased PRL after parturition when estrogen decreases abruptly in order to stimulate milk production.

Eating and Sleeping appear to generally stimulate (+) whereas Dehydration inhibits (-) PRL secretion. Mental and physical Stress responses are complicated and variable (+/-)

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In addition to anterior pituitary regulatory hormones, neurons (supraoptic & paraventricular nuclei) of the **Hypothalamus** (*top right*) produce (Hormone Synthesis) the hormones **Vasopressin** (Antidiuretic Hormone = **ADH**) & **Oxytocin (OT)**, which are transported (Axoplasmic Transport) to the **Posterior Pituitary Gland** where they are stored there until a secretion signal is received (Hormone Storage & Secretion).

Vasopressin mainly acts to stimulate **Renal H_2O Reabsorption** from the late distal tubules & collecting ducts (Antidiuretic effect). In addition, it has a potent **Vasoconstriction** effect on blood vessels in response to hypotension (pressor effect).

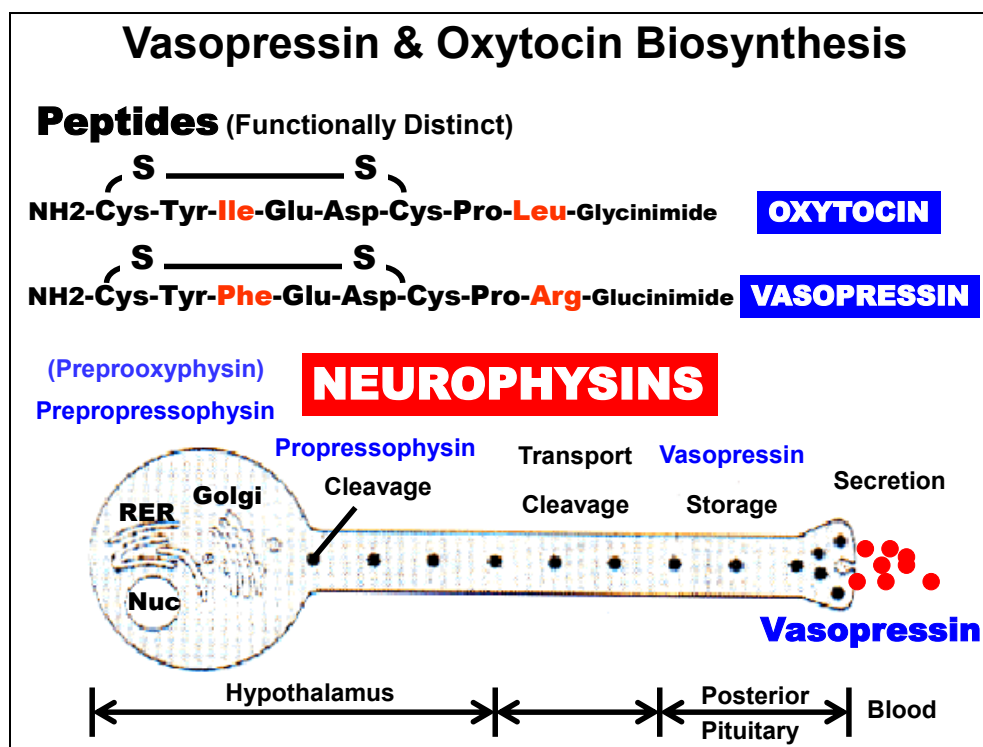
Oxytocin stimulates smooth muscle contraction. As mentioned earlier, oxytocin is responsible for **Milk Ejection** during **Lactation** in concert with milk production by **Prolactin**. In addition, oxytocin is thought to be involved in stimulating **Uterine Contractions** during child birth (**Labor**) although this effect may not be physiological. Discussed more during the Reproduction section of course.

Vasopressin & Oxytocin Properties

- **Classification:** **Proteins**
- **Biosynthesis:** **Hypothalamus (Neurophysins)**
- **Storage:** **Posterior Pituitary (Secretory Vesicles)**
- **Secretion:** **Exocytosis**
- **Transport:** **Dissolved**
- **Metabolism:** **Proteolysis (Half-Life: Minutes)**
- **Mechanism:** **Membrane Receptor: G-Protein**
- **Actions:** **ADH: Renal Tubules/Vascular SM**
Oxytocin: Mammary & Uterine SM
- **Regulation:** **Negative/Positive/Open-Loop Feedback**
- **Disorders:** **ADH: SIADH, Diabetes Insipidus**
Oxytocin: Lactation, Labor Complications

Vasopressin (ADH) and Oxytocin endocrine properties

Highlights are summarized above.



(top) **Vasopressin & Oxytocin** are both nine amino acid **Peptides Hormones** that differ at only two amino acid position that confer marked physiological functional distinction.

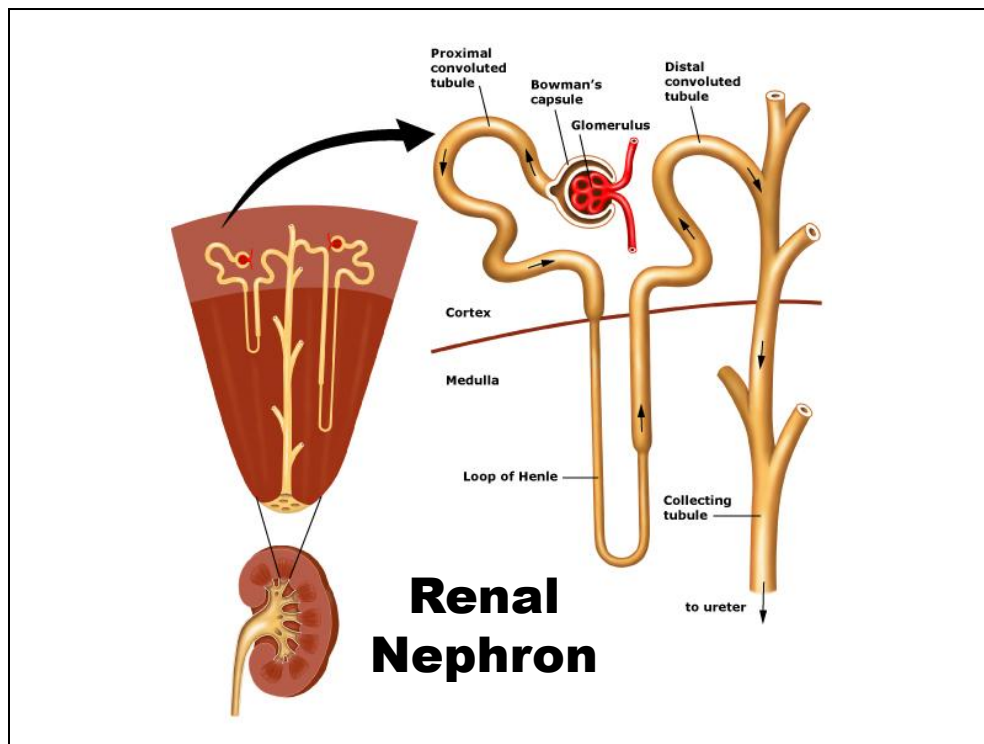
(bottom) Like most peptide hormones, vasopressin & oxytocin are synthesized initially as larger inactive prepropeptides that are generally referred to as **Neurophysins**, which were originally thought to be separate binding polypeptides but are, in fact, a part of the synthesized precursor molecules.

The precursor for vasopressin, **Prepropressophysin**, contains a leader sequence followed by the vasopressin peptide, neurophysin sequence and finally a glycoprotein moiety, which is absent from the oxytocin precursor, **Preprooxyphysin**.

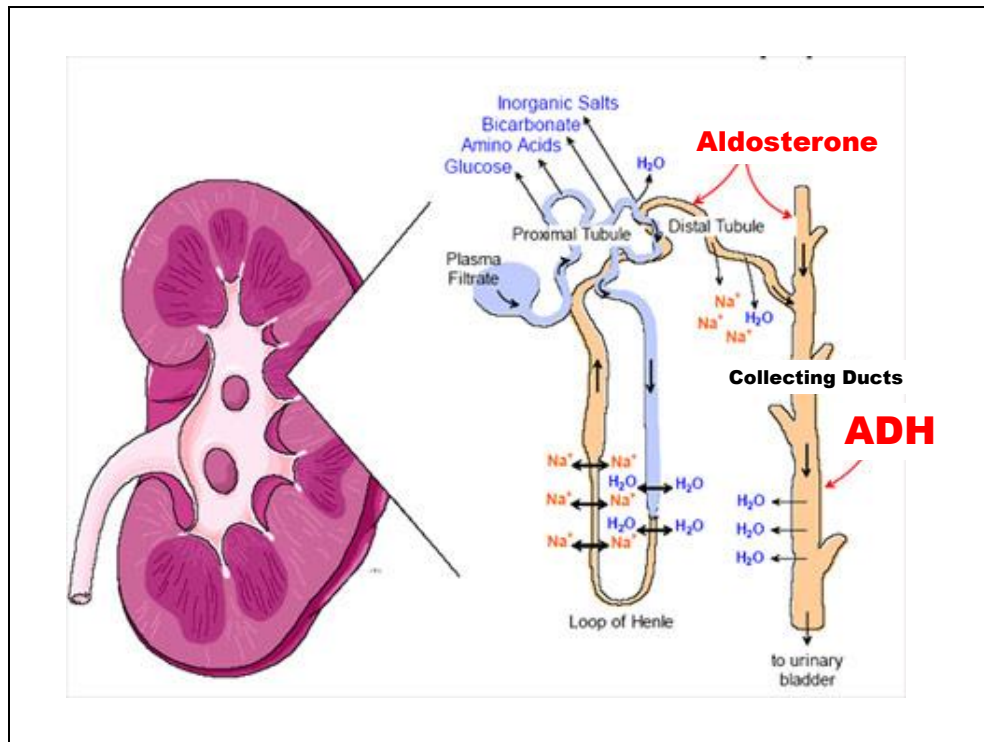
As shown, Cleavage of Prepropressophysin (similar for preprooxyphysin) is initiated shortly after synthesis in the Hypothalamus and is completed during Transport to the Posterior Pituitary for Storage of the mature hormones until Secretion is stimulated into Blood.

VASOPRESSIN

(Antidiuretic Hormone)

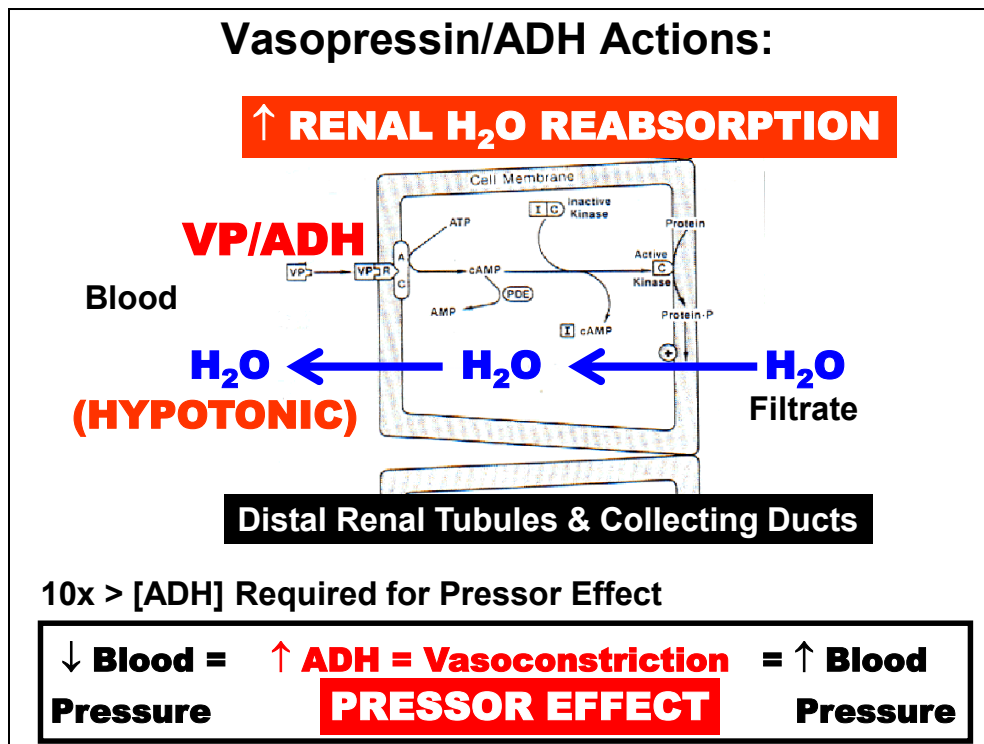


Reminder of functional anatomy of renal nephron. Hormones regulate dynamics at specific regions along tubule.



Aldosterone (discussed in Adrenal Gland section) and **ADH** regulate distinct mechanisms related to electrolyte and water reabsorption, respectively, along the renal tubule.

Thus, both are important hormones regulating blood pressure and electrolyte balance.



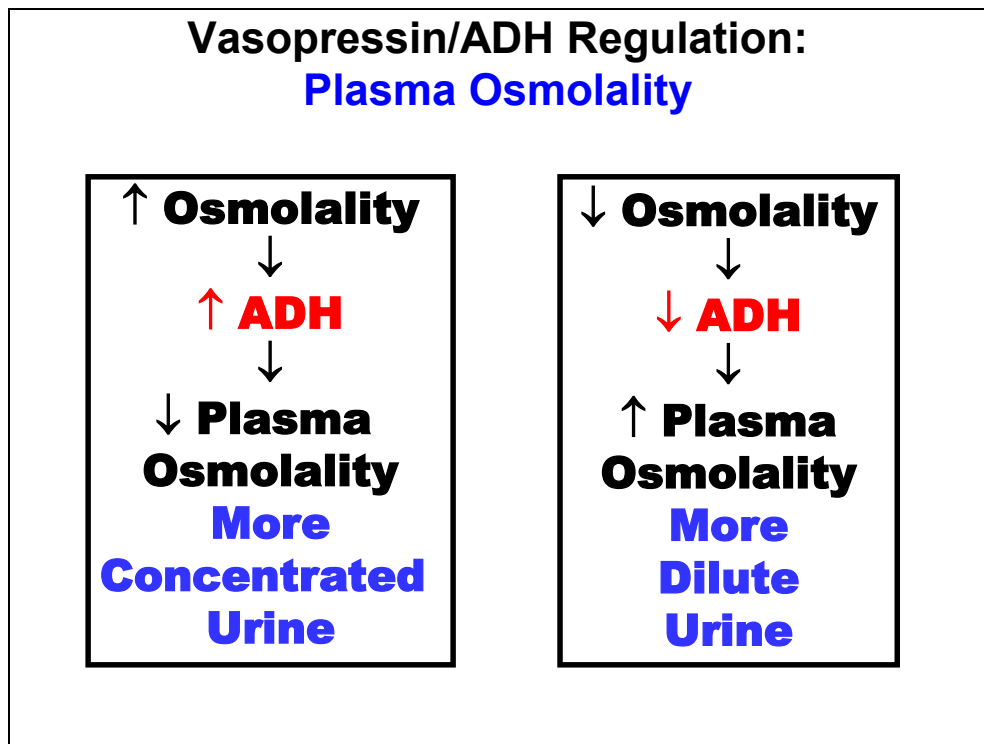
H₂O Reabsorption (Antidiuretic) Action. Regulation of renal H₂O reabsorption (late distal tubules & collecting ducts) is the primary physiological action of ADH. The cellular mechanism is shown schematically above.

Note **ADH** action results in **HYPOTONIC** fluid reabsorption. This is in contrast to **Aldosterone** (presented later in Adrenal Gland section), which causes reabsorption of an **Isotonic** fluid volume.

Pressor (Vasoconstrictor) Action. Vasopressin/ADH also stimulates **Vasoconstriction** of vascular smooth muscle. This pressor effect of ADH is most commonly elicited in response to a significant decrease (↓) in **Blood Pressure** e.g. severe **Hemorrhage** (*bottom*).

An important concept is that the **pressor response requires a plasma ADH concentration ≈10x that needed for the maximum Antidiuretic effect.**

Thus, ADH regulated pressor effects are **NOT** typically elicited within the normal ADH secretion range stimulated in response to physiological osmoregulation.



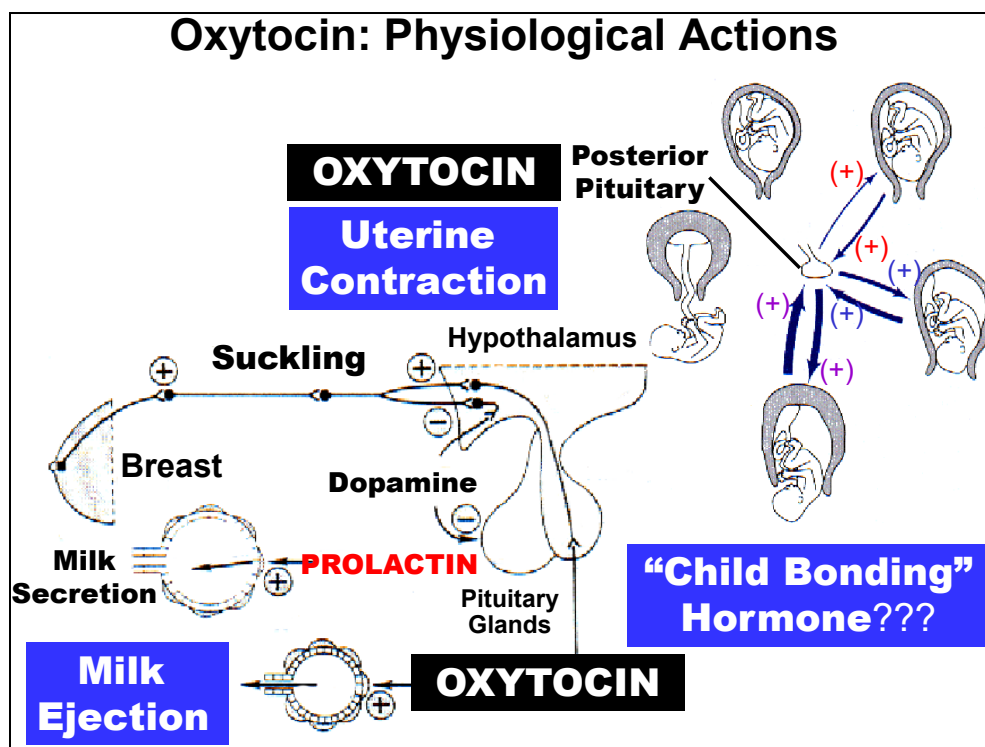
Plasma Osmolality is the primary physiological regulator of **ADH** biosynthesis and secretion.

The predicted response of ADH secretion to an increase (left) and decrease (right) in plasma osmolality is summarized along with the expected effect on plasma osmolality and urine concentration:

(↑) plasma **Osmolality** (↑) **ADH** secretion & H₂O reabsorption, which (↓) plasma **Osmolality** and results in the excretion of a **More CONCENTRATED Urine**.

(↓) plasma **Osmolality** (↓) **ADH** secretion & H₂O reabsorption, which (↑) plasma **Osmolality** and results in the excretion of a **More DILUTE Urine**.

OXYTOCIN



Oxytocin is a posterior pituitary hormone that regulates diverse reproductive processes.

Milk Ejection (Let-Down) during lactation is the principal physiologic function of **Oxytocin** (*lower left*). Oxytocin stimulates contraction of myoepithelial cells surrounding mammary alveoli, which squeezes milk from alveoli into large ducts that eventually lead out through the nipple. Oxytocin is NOT involved in milk production.

Uterine Contraction. Stimulation of **Uterine** (smooth muscle) **Contraction** was the first recognized oxytocin action. However, its physiological importance remains equivocal, the smooth muscle contracting action of oxytocin is routinely employed pharmacologically to induce **Labor** and manage obstetric **Hemorrhage** (vasoconstriction).

"Child Bonding" Hormone??? Oxytocin has been linked to certain psychosocial and sexual behaviors, including maternal bonding with newborns. In this context, regular breastfeeding (increased oxytocin secretion) is hypothesized as a mechanism by which mothers are physiologically compelled to care for their young.



Williams syndrome (WS) is a condition caused by a deletion of ~26–28 genes on chromosome 7q11.23 often characterized by abnormal social behavior and disrupted oxytocin (OT) and vasopressin (AVP) functioning.

The observation that individuals with WS exhibit OT and AVP dysregulation is compelling. Data from some studies indicate that the missing genes affect the release of oxytocin and AVP through the hypothalamus and the pituitary gland.

<https://www.salk.edu/news-release/trust-hormone-oxytocin-found-at-heart-of-rare-genetic-disorder/>

Williams Syndrome

What is Symptoms

- Causes: Dental problems
- Kidney problems
- Sensitive hearing
- Excessively friendly 
- Hernia Problems
- Distorted Facial features
- Troubles eating
- They have Low birth weight and low weight gain
- Intellectual Disability
- Respiratory Problems



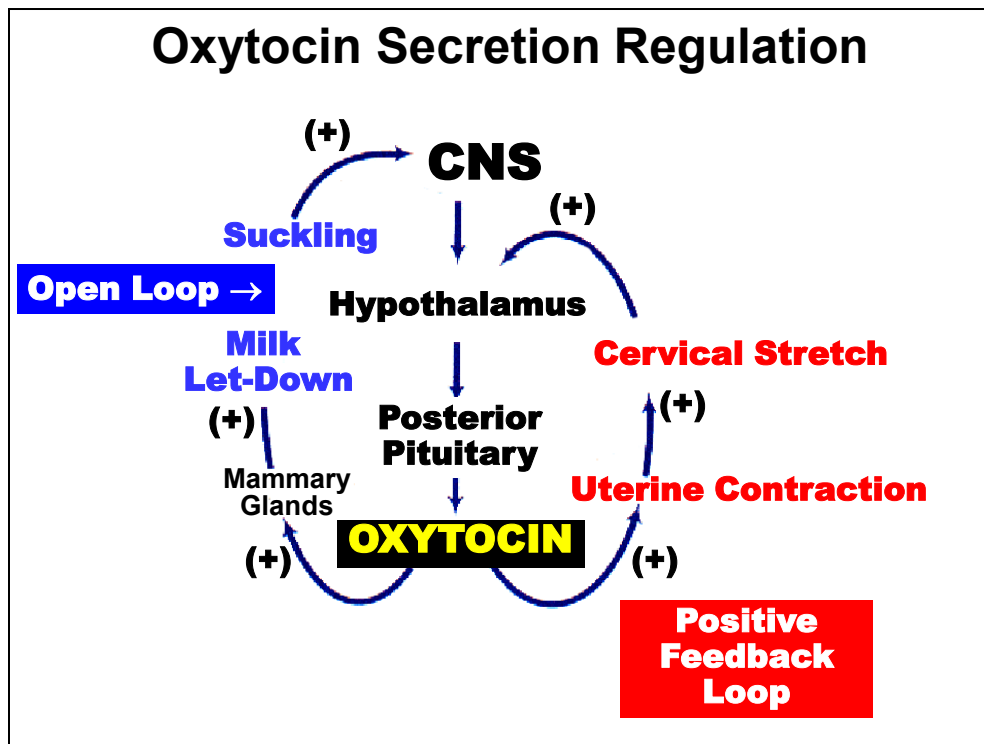
Joshua Buccheri

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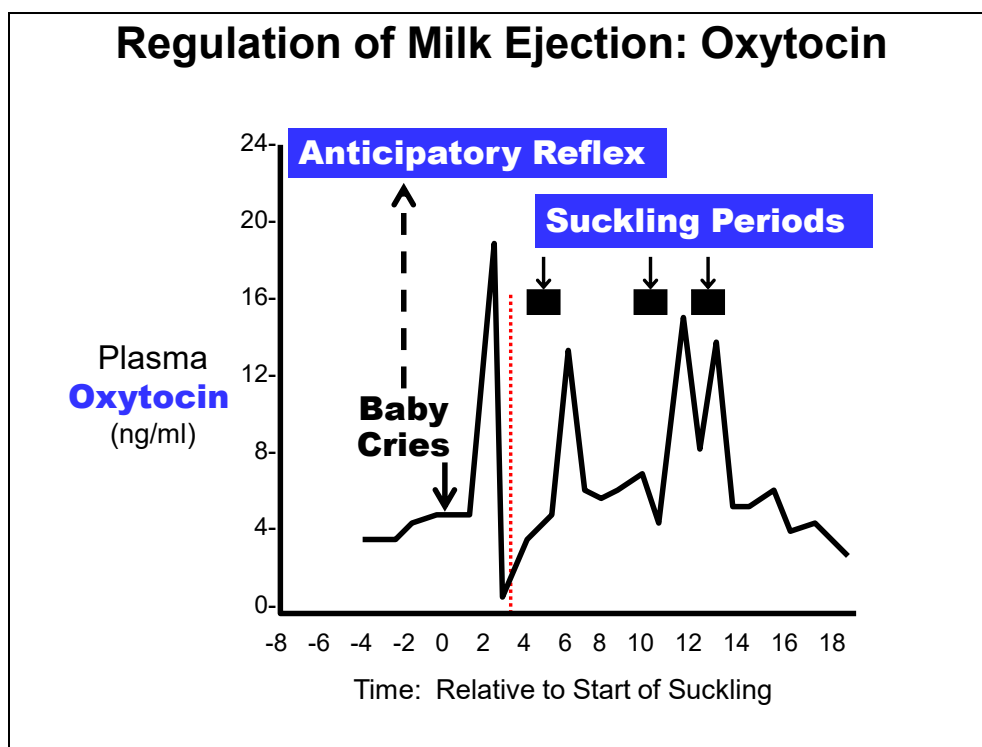
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(*right loop*). **Oxytocin** induced **Uterine Contraction** is one of only a few physiological **Positive Feedback Loops** used for hormone regulation. As diagrammed, **Cervical Stretch** is sensed by cervical neurons, which transmit afferent fibers to the Hypothalamus, signaling oxytocin secretion from the posterior pituitary. As the downward movement of the infant further stretches the cervix, oxytocin secretion is stimulated even more (+). This allows the development of sustained & powerful uterine contractions during labor.

(*left loop*). Like prolactin, oxytocin secretion stimulation by suckling is an example of **Open-Loop** hormone secretion control. Breast nipple **Suckling** initiates a stimulatory neural reflex transmitted from afferent sensory nerve endings in the nipple to the CNS and hypothalamus.

Oxytocin release is triggered from the posterior pituitary, exerting its contractile (+) action on **myoepithelial cells of Mammary Glands**, which results in **Milk Let-Down**. Stimulatory input is interrupted only when suckling stops.



Oxytocin levels (*y-axis*) increase during **Suckling Periods** (*shaded boxes*) throughout the day (*x-axis*) and decrease when suckling ceases. In some women, oxytocin secretion & milk ejection may be stimulated prior to nipple suckling via a conditioned reflex triggered when a hungry **Baby Cries** or when following a highly regimented feeding schedule.

This so-called **Anticipatory Reflex** is mediated through higher brain centers, but is not well understood.

By contrast, **Prolactin** secretion (*not shown*) is **NOT** apparently stimulated via this reflex mechanism.

Summary: Hypothalamic & Pituitary Hormones			
HYPOTHALAMIC HORMONE	PITUITARY HORMONE	TARGET ORGAN	TARGET ACTIONS
Corticotropin Releasing H. (CRH)	Adrenocorticotrophic H. (ACTH)	Adrenal Cortex	Glucocorticoids Androgen Secretion
Thyrotropin Releasing H. (TRH)	Thyroid Stimulating H. (TSH)	Thyroid Gland	Thyroxine; T3 Secretion
Gonadotropin Releasing H. (GnRH)	Follicle Stimulating H. (FSH) Leutinizng H. (LH)	Gonads	Estrogens; Progest. Testosterone Ovarian Cyc; Sperm
GH Releasing Hormone. (GHRH) (+) GH Release Inhibiting H. (Somatostatin: SST) (-)	Growth Hormone (GH)	Liver, Bone Muscle, Fat	IGF-1, Bone Muscle, Fat Metab.
Prolactin Releasing Factors (PRF's) (+) Prolactin Release Inhibit. H. (PRIH) (Dopamine: DA) (-)	Prolactin (PRL)	Breast	Milk Production & Secretion
Vasopressin (VP) or Antidiuretic H. (ADH)	Posterior Pituitary Storage & Secretion	Renal Tubules Blood Vessels	H ₂ O Reabsorption Vasoconstriction
Oxytocin (OT)	Posterior Pituitary Storage & Secretion	Breast Uterus	Milk Let-Down Labor Contractions

Summary of hypothalamic and pituitary hormones.

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